Two-Equation Turbulence Models and Effect of Compressibility

Problem

The objective of this study is to model a subsonic jet over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the SST model with different inlet turbulence boundary conditions (Turbulent intensity 1%, 5% and 15%). This simulation will validate the CFD results against experimental results for:

- profiles of $u'v'/U_{jet}^2$, u/U_{jet}, v/U_{jet} and k/U_{jet}^2 at $x/D_{jet} = 2$, 5, 10, 15 and 20.

This study helps assess the influence of varying inlet turbulence intensity on the accuracy of jet flow predictions using the SST model by comparing key turbulence and velocity profiles with experimental data.

Solution

Subsonic jet flows are common in numerous engineering applications, including propulsion, HVAC systems, and industrial mixing processes. Accurate prediction of such flows is crucial, particularly in the near-field region where turbulence strongly influences the mixing characteristics. Computational Fluid Dynamics (CFD) provides a powerful tool for investigating jet behavior and validating turbulence models against experimental data.

This study investigates a subsonic round jet discharging from a nozzle over a flat plate, modeled using ANSYS Fluent. The axisymmetric jet configuration is representative of the ARN1 (Acoustic Research Nozzle 1, or ARN1) experimental setup. The primary objective is to assess the performance of the Shear Stress Transport (SST) k–ω turbulence model under varying inlet turbulence intensities (1%, 5%, and 15%). The simulation results are compared with experimental data to evaluate their accuracy, particularly in terms of normalized velocity and turbulence profiles at selected streamwise locations.

Setup

The computational study was performed using ANSYS Fluent in a 2D axisymmetric configuration to model a subsonic jet issuing from a nozzle over a flat plate. The mesh domain represents a 1-degree slice rotated about the x-axis, with the jet centerline lying along the axis of symmetry ($y = 0$), presented on the Fig. 1. The simulation employed a steady-state, pressure-based solver. The velocity formulation was set to absolute, and the simulation was carried out under the assumption of axisymmetry to exploit geometric simplicity while capturing radial flow features.

Figure 1 shows the boundary conditions applied in the domain:

- **Inlet jet:** The jet inflow condition was applied with $P/P_{ref} = 1.19671$ and $T_t/T_{ref} = 1.0$ at $x = 0$, with meaning that initial supersonic and total gauge pressure were 119671 Pa at the inlet and temperature 295 K. These reference conditions are consistent with the ARN1 axisymmetric jet experiment. Turbulent intensity was set to 1%, 5%, and 15% for different cases.
- **Inlet freestream:** Velocity inlet with 3.43 m/s axial velocity was specified. The inlet was located upstream of the nozzle exit to allow full development of the jet.
- **Wall:** The nozzle surface was treated as an adiabatic solid wall, enforcing a no-slip condition and zero heat flux.

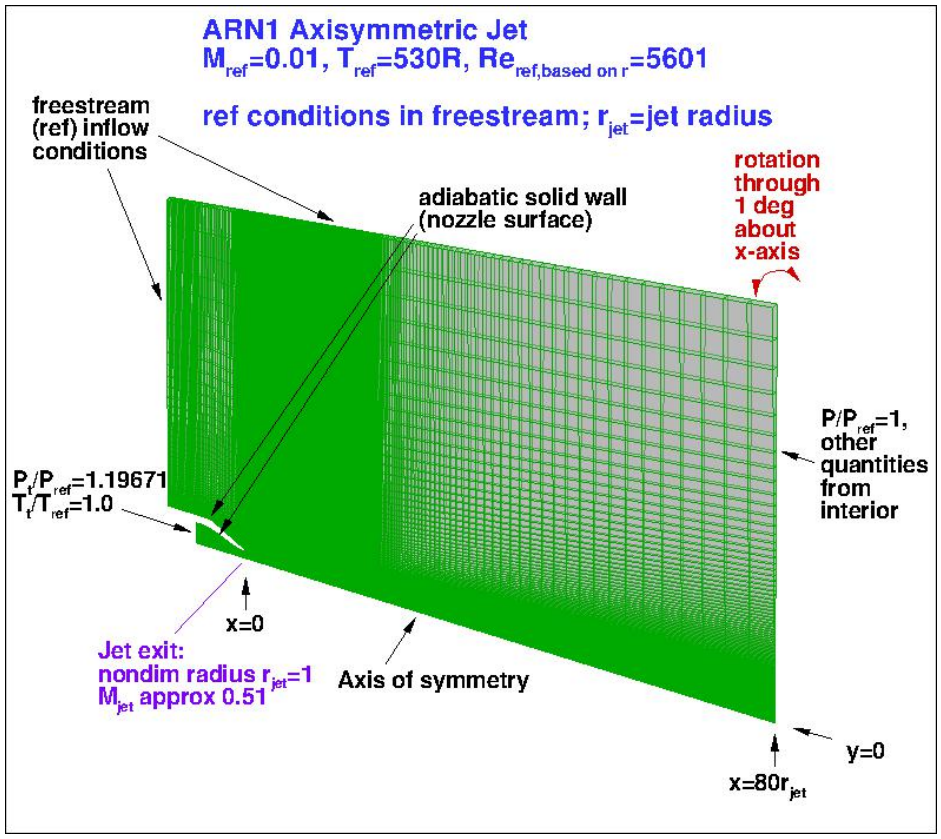

Figure 1: Computational domain and boundary conditions for the axisymmetric jet simulation, taken from [1].

- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet ($x = 80r_{jet}$), a pressure outlet condition was applied with $P/P_{ref} = 1$. Set as a pressure outlet with the option *prevent reverse flow* enabled to avoid non-physical backflow and enhance solution stability.

The SST (Shear Stress Transport) k–ω model was selected for turbulence modeling. Three different inlet turbulence intensities—1%, 5%, and 15%—were tested to examine their influence on the jet's development and agreement with experimental data.

Postprocessing was performed in Ansys Fluent and Python to extract relevant flow quantities along radial profiles at several downstream locations from the jet exit. The primary objective was to evaluate the influence of inlet turbulence intensity on the accuracy of predicted velocity and turbulence characteristics by comparing simulation results with experimental data.
The following non-dimensional quantities were extracted and plotted at $x/D_{jet} = 2$, 5, 10, 15, and 20:

- Normalized axial velocity: $\frac{u}{U_{jet}}$
- Normalized radial velocity: $\frac{v}{U_{jet}}$
- Normalized Reynolds shear stress: $\frac{u'v'}{U_{jet}^2}$

- Normalized turbulent kinetic energy: $\frac{k}{U_{jet}^2}$

The jet exit velocity U_{jet} was determined from the simulation by taking the value of the velocity magnitude along the centerline (axis of symmetry) at the nozzle exit ($x = 0$). The maximum value of the velocity magnitude at this location was taken as U_{jet}. The quantities were calculated using custom-defined field functions in Fluent as follows:

$$\frac{u}{U_{jet}} = \frac{\text{axial-velocity}}{U_{jet}} \tag{1}$$

$$\frac{v}{U_{jet}} = \frac{\text{radial-velocity}}{U_{jet}} \tag{2}$$

$$\frac{k}{U_{jet}^2} = \frac{\text{turb-kinetic-energy}}{U_{jet}^2} \tag{3}$$

$$\frac{u'v'}{U_{jet}^2} = \frac{-\mu_t}{\rho}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\frac{1}{U_{jet}^2} \tag{4}$$

The results from the CFD simulations were plotted alongside experimental data provided in a structured dataset, allowing direct comparison at each downstream location. This comparison aids in evaluating the predictive performance of the SST model under varying inlet turbulence intensities.

Shear-Stress Transport (SST) $k - \omega$

The SST $k - \omega$ model accounts for the transport of the turbulence shear stress in the definition of the turbulent viscosity. These features make the SST model more accurate and reliable for a wider class of flows (for example, adverse pressure gradient flows, airfoils, transonic shock waves) than the above mentioned standard models [2]. Other models fail to properly predict the onset and amount of flow separation from smooth surfaces. The main reason is that other models do not account for the transport of the turbulent shear stress. This results in an overprediction of the eddy-viscosity. The proper transport behavior can be obtained by a limiter to the formulation of the eddy-viscosity:

$$\mu_t = \frac{\rho k}{\omega}\frac{1}{max[\frac{1}{\alpha^*}, \frac{SF_2}{a_1\omega}]}, \tag{5}$$

where S is the strain rate magnitude and α^* is defined as

$$\alpha^* = \alpha_\infty^*\left(\frac{\alpha_0^* + Re_t/R_k}{1 + Re_t/R_k}\right), \quad Re_t = \frac{\rho k}{\mu\omega}, \quad R_k = 6, \quad \alpha_0^* = \frac{0.072}{3}. \tag{6}$$

F_2 is given by

$$F_2 = \tanh \Phi_2^2, \qquad \Phi_2 = max\left[2\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2\omega}\right], \tag{7}$$

where y is the distance to the next surface.

Results & Discussion

The Mach number distribution at the outlet was examined to confirm the subsonic nature of the jet, Fig. 2. The maximum Mach number obtained was approximately 0.525, indicating that the flow remains within the subsonic regime. This validates the use of a pressure-based solver, which is appropriate for low-speed, incompressible or mildly compressible flows.

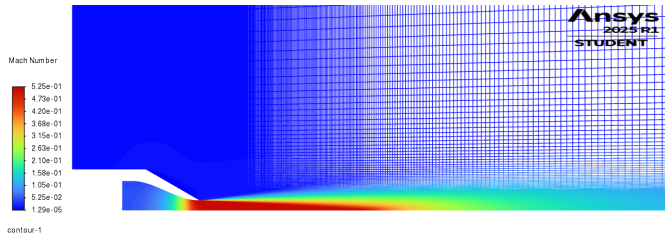

Figure 2: Mach number distribution at the outlet for 5% jet inlet turbulence intensity.

Simulation result of u/U_{jet} axial velocity distribution is plotted on Fig. 3. The axial velocity profiles at downstream positions $x/D_{jet} = 2$, 5, 10, 15, and 20 show a progressive broadening of the jet with increasing x/D_{jet}, indicative of jet spreading and momentum diffusion into the surrounding fluid. At $x/D_{jet} = 2$, the velocity profile is still relatively sharp, resembling a top-hat shape characteristic of a potential core region. As the flow evolves downstream, the profiles transition toward a self-similar Gaussian-like shape, with reduced peak velocity and increased radial spread.

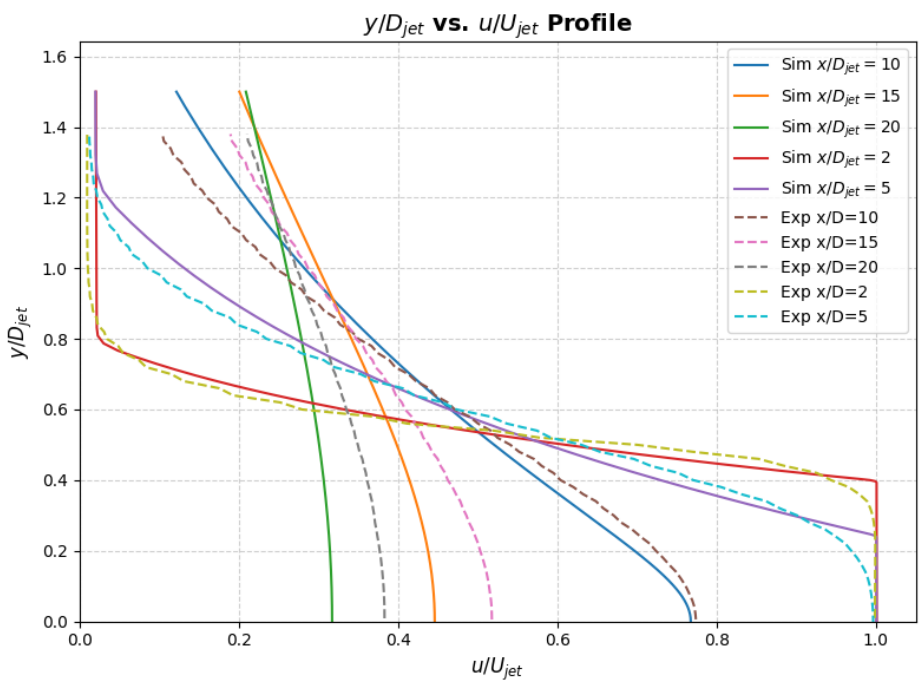

Figure 3: Axial velocity profile distribution for 5% jet inlet turbulence intensity.

Interestingly, across all tested turbulence intensities (1%, 5%, and 15%), the velocity profiles were nearly identical at all streamwise locations. This indicates that the SST $k-\omega$ model, within the context of this simulation, is relatively insensitive to variations in inlet turbulence intensity. The turbulent mixing and shear layer development appear to be primarily governed by the flow dynamics rather than the specified inlet turbulence level. In the near field ($x/D_{jet} = 2$), the CFD results show good agreement with experimental data, both in the shape and magnitude of the

velocity profile. This suggests the SST $k-\omega$ model effectively captures the initial development of the jet shear layer.

However, at larger downstream positions ($x/D_{jet} \geq 5$), noticeable discrepancies begin to emerge. The simulation under-predicts the axial velocity, particularly near the centerline region ($y/D_{jet} \approx 0$), where the experimental data show a higher velocity peak. This underestimation becomes more pronounced as x/D_{jet} increases, indicating that the model fails to sustain the core velocity as effectively as observed in the experiment.

The insensitivity of the SST model to inlet turbulence intensity—evidenced by nearly identical results for 1%, 5%, and 15% inlet turbulence—may contribute to this deviation. While the model self-generates turbulence and adequately captures the initial mixing, it may not correctly model the long-range momentum diffusion or turbulence decay processes that influence far-field jet behavior. These results suggest that while the SST $k-\omega$ model is suitable for near-field jet predictions, its accuracy degrades further downstream, particularly in replicating the centerline velocity distribution. This highlights a potential limitation in the model's predictive capability for fully developed jet flow unless more refined turbulence modeling or transition specification is introduced.

Simulation result of v/U_{jet} radial velocity distribution is presented on Fig. 4. The normalized radial velocity profiles provide insight into the entrainment behavior of the jet as it develops downstream. In contrast to axial velocity, which is dominant along the centerline, radial velocity peaks away from the centerline and reflects the outward movement of fluid driven by shear layer expansion and turbulent mixing.

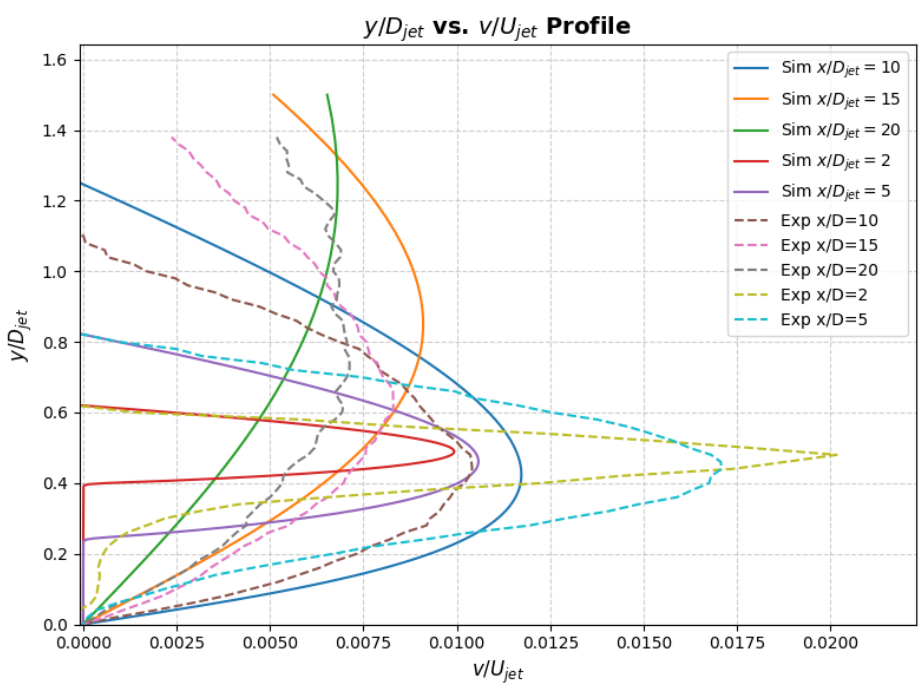

Figure 4: Radial velocity profile distribution for 5% jet inlet turbulence intensity.

Across all streamwise locations, the radial velocity profiles from the simulation exhibit the characteristic behavior of a turbulent jet: low values near the centerline ($y/D_{jet} \approx 0$), with a gradual rise to a peak at intermediate (y/D_{jet}), followed by a decay toward the outer region. This peak corresponds to the location of the shear layer, where turbulent mixing is most intense.

However, when compared to the experimental data, the simulated radial velocities are generally underpredicted at most locations, particularly in the far field. This underestimation is consistent with the axial velocity discrepancies, suggesting a general trend where the model does not fully capture the extent of turbulent mixing and momentum transfer in the outer regions of the jet. The shape of the profile is similar to the experimental curves, but the magnitude of v/U_{jet} is lower. An exception occurs at x/D_{jet}, where the simulation overpredicts the radial velocity, especially in the region near the peak ($y/D_{jet} \approx 1$). This overprediction could be due to a localized amplification of shear layer activity in the model, possibly related to how the SST model balances the k and ω transport near that axial location. It's also possible that numerical diffusion or mesh resolution in that region influenced the velocity gradients and contributed to the spike.

Moreover, similar to the axial velocity results, the radial velocity profiles show negligible sensitivity to the inlet turbulence intensity. The fact that 1%, 5%, and 15% turbulence intensity cases produce nearly identical results implies that the SST model generates internal turbulence rapidly, overshadowing the effects of prescribed inlet conditions. Overall, while the SST model qualitatively reproduces the structure of the radial velocity profiles, it appears to underestimate the strength of lateral entrainment, particularly further downstream. This could be a consequence of limited turbulent diffusion or energy transfer in the model, again pointing to possible improvements through higher-fidelity turbulence models or better inlet turbulence specification.

Simulation of Reynolds shear stress distribution $u'v'/U_{jet}^2$ is presented on Fig. 5. The Reynolds shear stress profiles provide a direct measure of turbulent momentum transport across the jet, driven primarily by velocity fluctuations in the axial and radial directions. In typical turbulent jet flows, $u'v'/U_{jet}$ exhibits a peak near the shear layer and approaches zero near the centerline and at large radial distances, consistent with reduced turbulence activity in these regions.

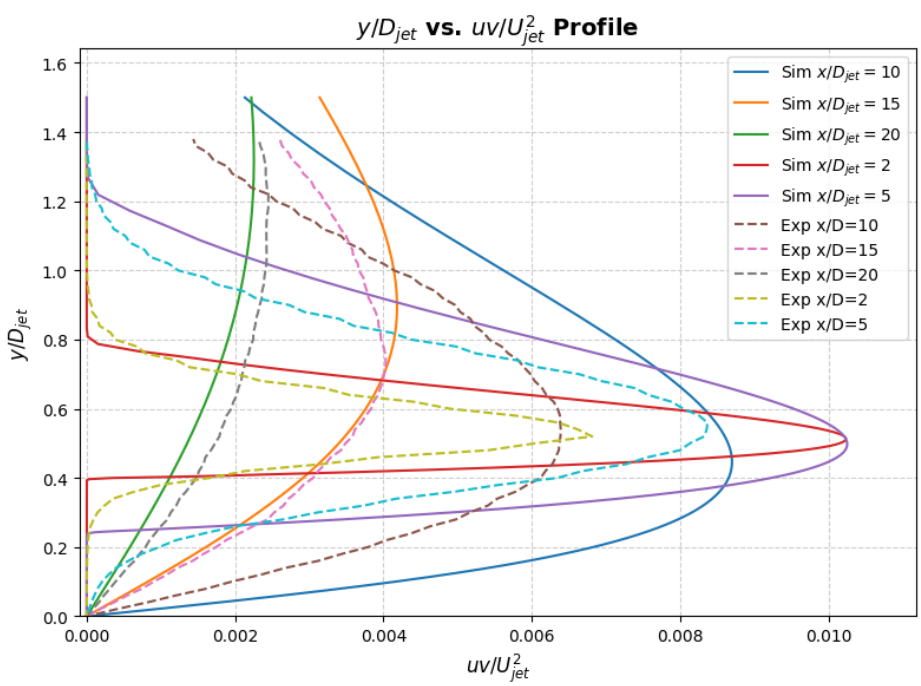

Figure 5: Reynolds shear stress distribution for 5% jet inlet turbulence intensity.

The simulation results show qualitatively correct behavior: at all streamwise locations, the Reynolds shear stress peaks at intermediate y/D_{jet}, near the edge of the potential core, and gradually diminishes toward the jet axis and far field. The location of the peak shifts outward with increasing x/D_{jet}, reflecting the radial growth of the turbulent shear layer. The shape of the distribution is very similar to that of the radial velocity, highlighting the physical link between turbulent mixing and lateral momentum transport.

However, the quantitative agreement with experimental data varies significantly across axial locations:

- At x/D_{jet}=2, 5, and 10, the simulation overpredicts the Reynolds shear stress throughout most of the shear layer. This suggests that the SST $k-\omega$ model generates stronger-than-observed turbulent momentum transport in the near and mid-field regions, likely due to its formulation being aggressive in capturing early shear layer development.
- At x/D_{jet}=15, the simulation underpredicts the shear stress for $y/D_{jet} < 0.7$, especially near the centerline. Beyond that point, the simulation begins to overpredict again in the outer shear layer. This reversal may point to a shift in the model's estimation of eddy viscosity or turbulence production away from the jet core.
- At x/D_{jet}=20, the model consistently underpredicts the Reynolds shear stress across the entire radial range. This underestimation aligns with similar trends seen in axial and radial velocity at far downstream locations, indicating that the SST model struggles to sustain turbulence intensity and mixing as the jet becomes more diffused.

Again, across all turbulence intensity cases (1%, 5%, 15%), the profiles remain effectively unchanged, reinforcing the earlier observation that inlet turbulence intensity has little influence on the predicted turbulence fields in the SST model for this configuration. The turbulence is likely generated primarily by the flow gradients at the nozzle exit and in the shear layer, overpowering the effect of the specified inlet conditions. In summary, while the SST $k-\omega$ model reproduces the general shape and location of the Reynolds shear stress profiles, it systematically underpredicts the magnitude of turbulent momentum transport when compared to experimental data, with limited sensitivity to inlet turbulence levels.

Simulation result of turbulence kinetic energy k is given on Fig. 6. The distribution of turbulent kinetic energy provides direct insight into the turbulence intensity and how energy is partitioned among fluctuating velocity components across the jet.

In all examined downstream locations, the simulation captures the expected qualitative features: low TKE along the centerline (=0), with a peak occurring within the shear layer, followed by a decline toward the jet's outer edge. This reflects the physical reality that turbulence is primarily generated in regions with high velocity gradients, especially at the boundary between the jet core and the ambient fluid.

However, the quantitative behavior of the simulation relative to experimental data varies with axial position:

- For x/D_{jet} = 2, 5, 10, 15 simulation overpredicts the turbulent kinetic energy across much of the radial profile. This overprediction aligns with earlier observations from the Reynolds shear stress distribution, where stronger turbulent mixing was also simulated in the near and mid-field regions.
- For x/D_{jet} = 20: The simulation matches the experimental turbulent kinetic energy throughout the radial profile.

The simulation results also show that inlet turbulence intensity has negligible effect on the predicted TKE profiles. All three cases (1%, 5%, and 15%) collapse onto nearly identical distributions, reinforcing that the model-generated turbulence dominates over the specified inlet conditions. In summary, while the SST $k-\omega$ model captures the general structure of the TKE distribution, it tends to overpredict turbulence levels in the near and mid-field and

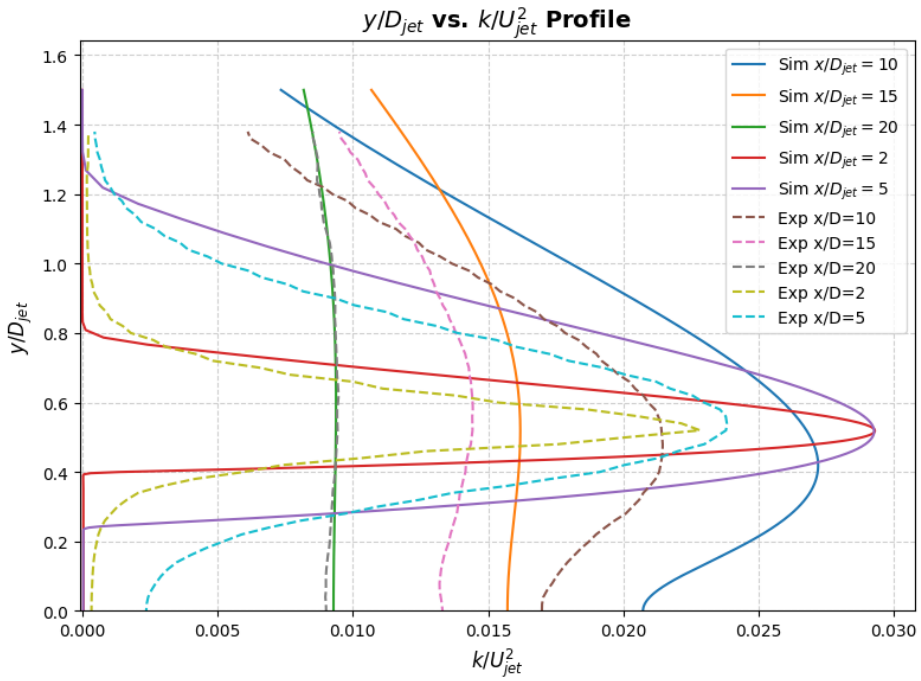

Figure 6: Turbulence kinetic energy distribution for 5% jet inlet turbulence intensity.

underpredict them in the far field, following a similar pattern observed in the Reynolds shear stress. These trends highlight the model's tendency to generate excessive turbulence early on, while potentially over-dissipating it farther downstream.

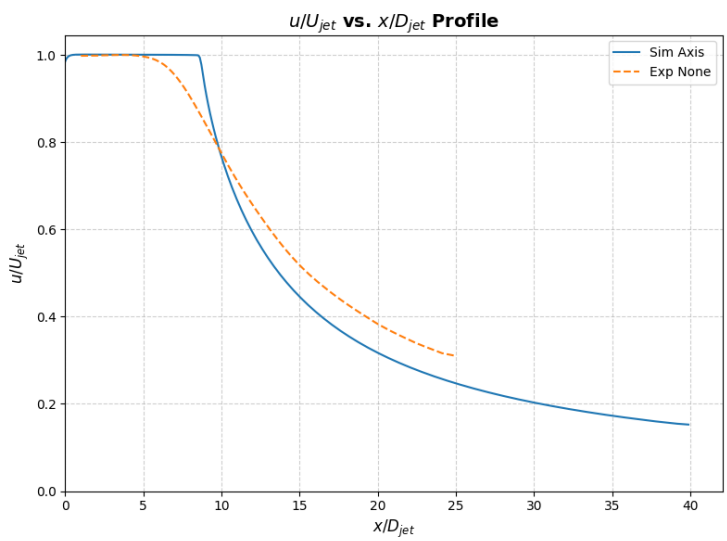

Figure 7: Axial velocity profile over axis for 5% jet inlet turbulence intensity.

The axial velocity profile u/U_{jet} decreases as the jet moves downstream, following typical jet development behavior. In the near-field (x/D_{jet}= 2,5), the simulation provides a reasonable match to experimental data, with high velocity near the centerline. In the far-field (x/D_{jet}=15, 20), discrepancies arise as the simulation tends to underpredict the velocity, possibly due to limitations in the turbulence model's ability to capture the full extent of jet expansion and dissipation. Overall, the distribution reflects the jet's natural tendency to lose momentum and spread radially with distance.

Conclusion

The analysis of the jet flow using the SST $k-\omega$ turbulence model reveals distinct trends in the predicted flow quantities compared to experimental data across various downstream locations. The simulation generally underpredicts the normalized axial velocity in the far-field regions (x/D_{jet}=15 and x/D_{jet}=20), particularly for smaller radial distances (x/D_{jet} <0.7), suggesting the model struggles with capturing the full extent of jet spreading and development. However, for closer axial locations (x/D_{jet}=2, 5, 10), the model provides a reasonable match to the experimental data.

The distribution of normalized radial velocity follows a similar pattern, with overprediction in the near-field (x/D_{jet}=2, 5, 10) and underprediction in the far-field (x/D_{jet}=15, 20), reflecting the model's difficulty in sustaining realistic turbulent mixing as the jet develops downstream. Notably, for x/D_{jet}=15, the simulation overpredicts the radial velocity in the outer shear layer, indicating potential numerical or model-driven overestimation of turbulence at this location. The Reynolds shear stress distribution exhibits similar trends to the radial velocity, with overpredictions in the near-field and underpredictions in the far-field. The turbulent kinetic energy profile shows overprediction in the near and mid-field regions (x/D_{jet} = 2, 5, 10) and good prediction in the far field (x/D_{jet}=20). This trend mirrors the behavior observed in Reynolds shear stress and radial velocity, reinforcing the idea that the SST model generates excessive turbulence in the early stages of jet development but fails to maintain accurate turbulence levels downstream.

Across all distributions, inlet turbulence intensity had minimal influence on the results, indicating that the SST model predominantly generates its own turbulence characteristics, with less reliance on the imposed inlet conditions. In conclusion, the SST $k-\omega$ model is effective for capturing the qualitative features of the jet flow, but it tends to overpredict turbulence in the near field and underpredict it in the far field, highlighting limitations in its ability to accurately model the jet's turbulent mixing and development over the entire domain.

References

[1] NASA Langley Research Center. Axisymmetric subsonic jet, 2021.

[2] Inc. ANSYS. *SST k-omega Model*, 2025.